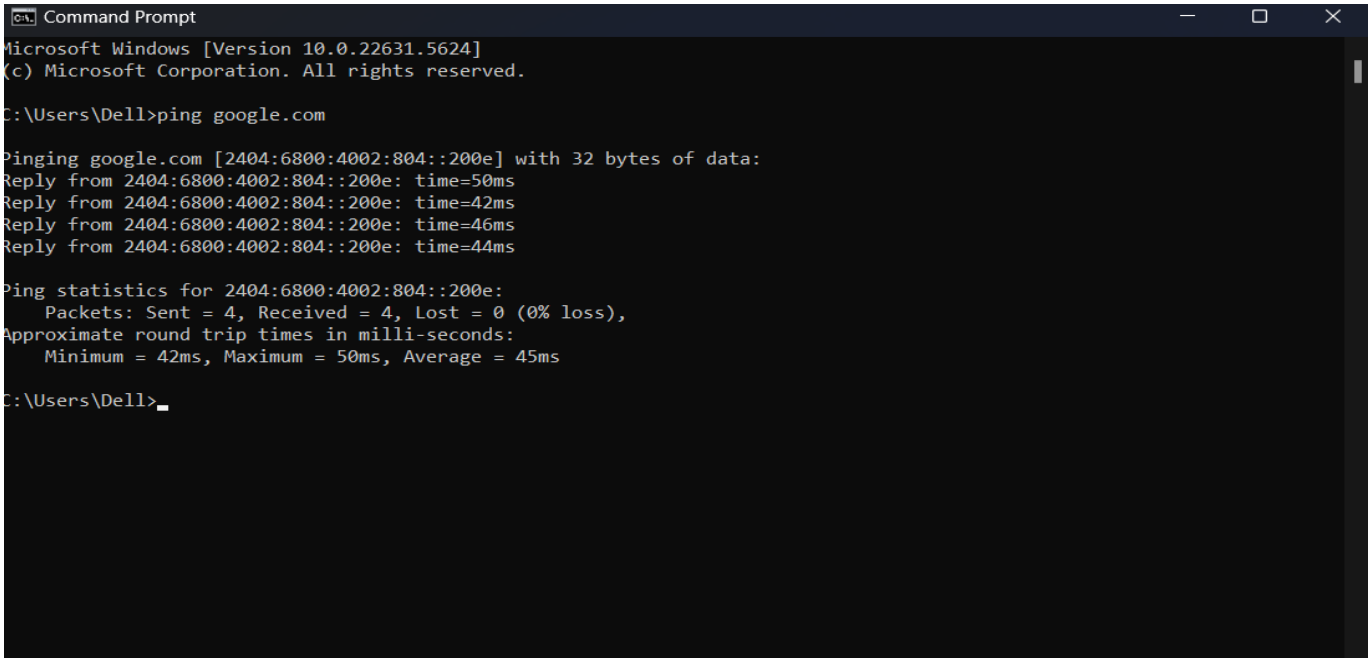
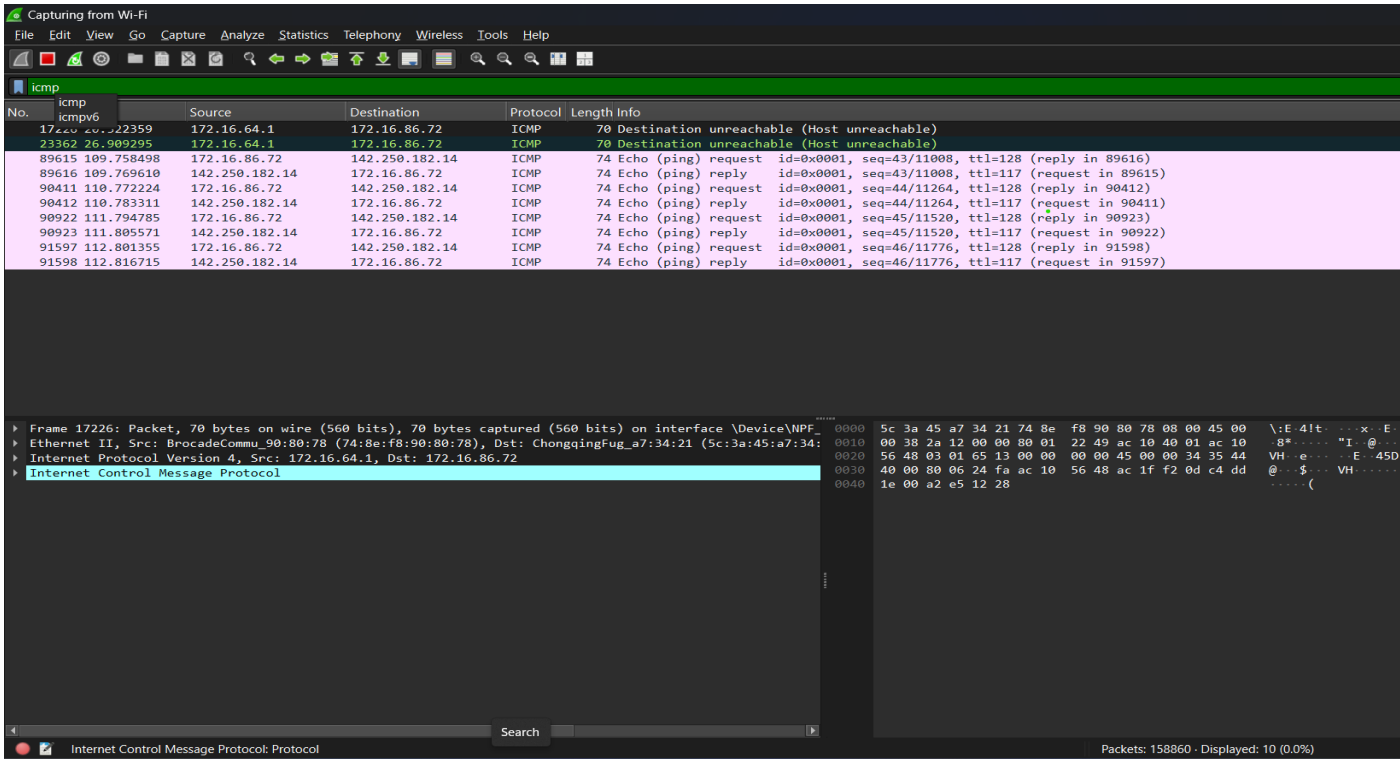
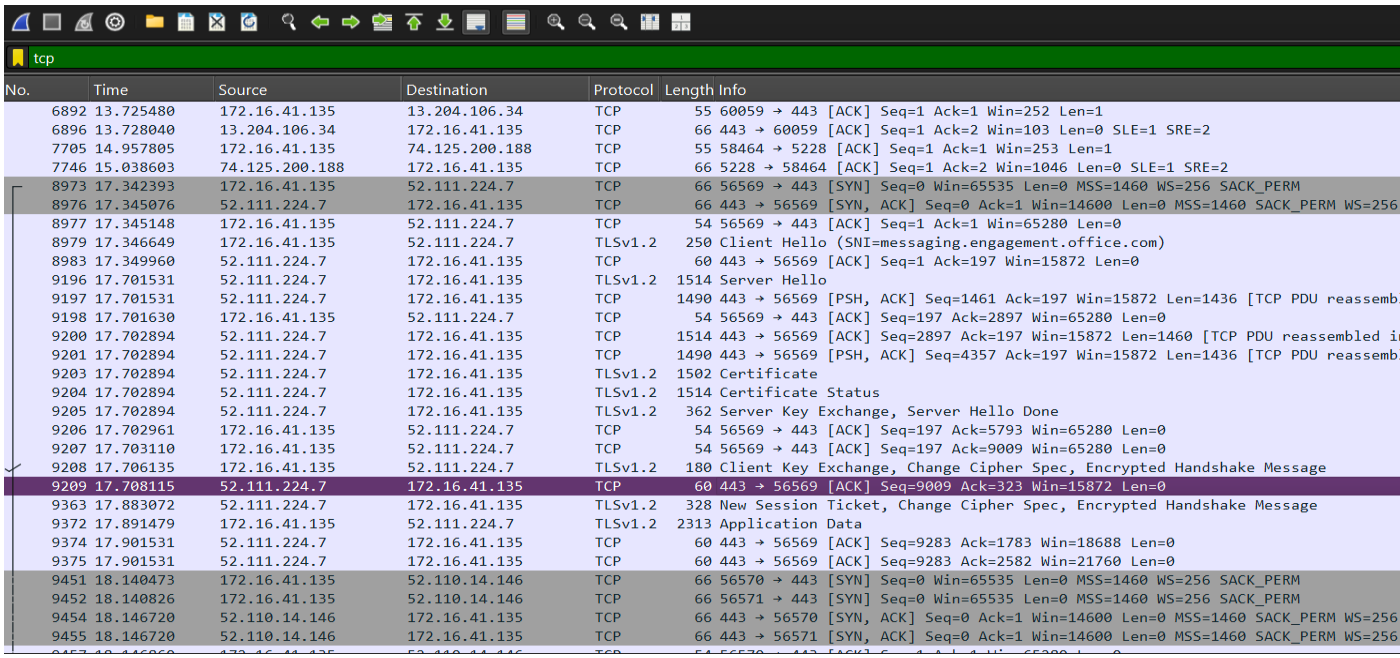
CN ASSIGNMENT -2

1. To capture and analyze ICMP packets (Ping request and reply)



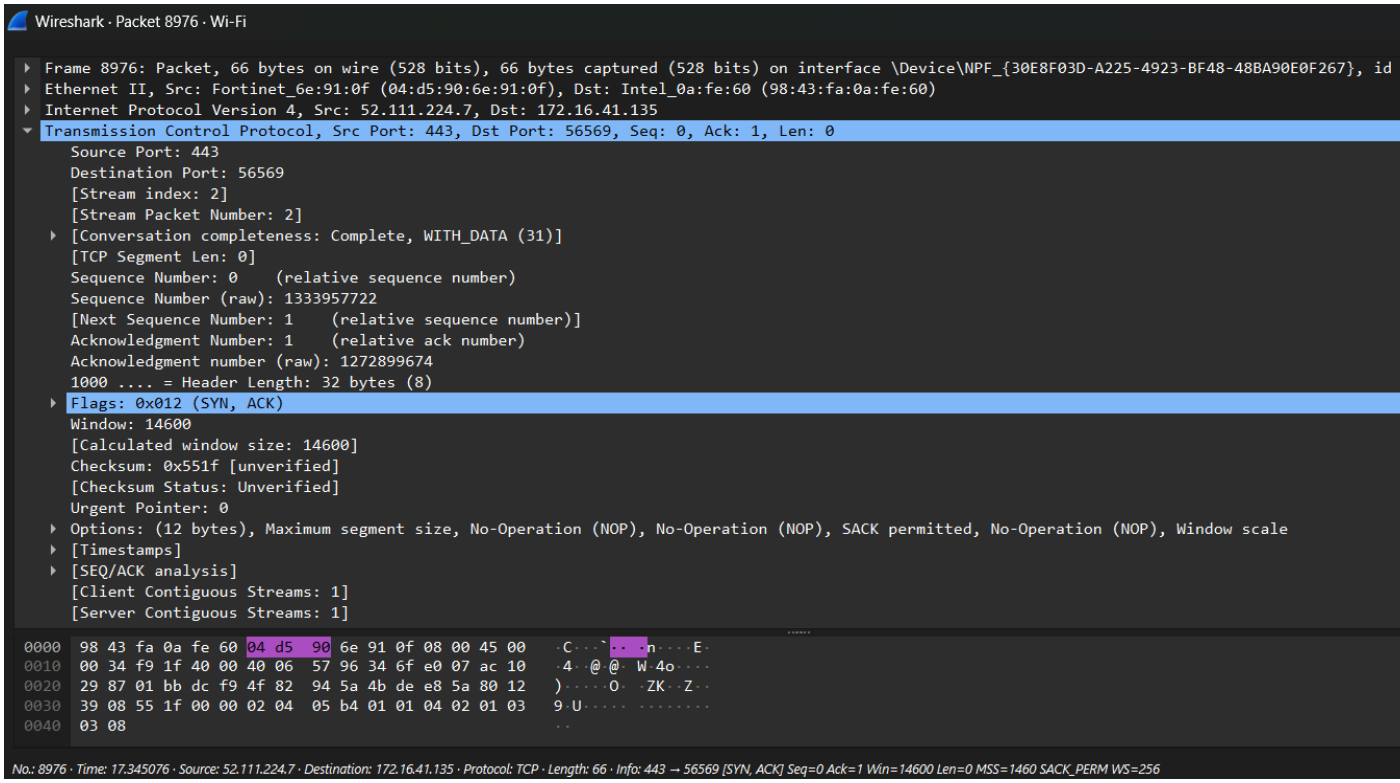


2. To analyze TCP packets and observe the TCP three-way handshake



tcp

No.	Time	Source	Destination	Protocol	Length	Info
6892	13.725480	172.16.41.135	13.204.106.34	TCP	55	60059 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
6896	13.728040	13.204.106.34	172.16.41.135	TCP	66	443 → 60059 [ACK] Seq=1 Ack=2 Win=103 Len=0 SLE=1 SRE=2
7705	14.957805	172.16.41.135	74.125.200.188	TCP	55	58464 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1
7746	15.038603	74.125.200.188	172.16.41.135	TCP	66	5228 → 58464 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
8973	17.342393	172.16.41.135	52.111.224.7	TCP	66	56569 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8976	17.345076	52.111.224.7	172.16.41.135	TCP	66	443 → 56569 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256
8977	17.345148	172.16.41.135	52.111.224.7	TCP	54	56569 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
8979	17.346649	172.16.41.135	52.111.224.7	TLSv1.2	250	Client Hello (SNI=messaging.engagement.office.com)
8983	17.349960	52.111.224.7	172.16.41.135	TCP	60	443 → 56569 [ACK] Seq=1 Ack=197 Win=15872 Len=0
9196	17.701531	52.111.224.7	172.16.41.135	TLSv1.2	1514	Server Hello
9197	17.701531	52.111.224.7	172.16.41.135	TCP	1490	443 → 56569 [PSH, ACK] Seq=1461 Ack=197 Win=15872 Len=1436 [TCP PDU reassemb
9198	17.701630	172.16.41.135	52.111.224.7	TCP	54	56569 → 443 [ACK] Seq=197 Ack=2897 Win=65280 Len=0
9200	17.702894	52.111.224.7	172.16.41.135	TCP	1514	443 → 56569 [ACK] Seq=2897 Ack=197 Win=15872 Len=1460 [TCP PDU reassembled i
9201	17.702894	52.111.224.7	172.16.41.135	TCP	1490	443 → 56569 [PSH, ACK] Seq=4357 Ack=197 Win=15872 Len=1436 [TCP PDU reassemb
9203	17.702894	52.111.224.7	172.16.41.135	TLSv1.2	1502	Certificate
9204	17.702894	52.111.224.7	172.16.41.135	TLSv1.2	1514	Certificate Status
9205	17.702894	52.111.224.7	172.16.41.135	TLSv1.2	362	Server Key Exchange, Server Hello Done
9206	17.702961	172.16.41.135	52.111.224.7	TCP	54	56569 → 443 [ACK] Seq=197 Ack=5793 Win=65280 Len=0
9207	17.703110	172.16.41.135	52.111.224.7	TCP	54	56569 → 443 [ACK] Seq=197 Ack=9009 Win=65280 Len=0
9208	17.706135	172.16.41.135	52.111.224.7	TLSv1.2	180	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
9209	17.708115	52.111.224.7	172.16.41.135	TCP	60	443 → 56569 [ACK] Seq=9009 Ack=323 Win=15872 Len=0
9363	17.883072	52.111.224.7	172.16.41.135	TLSv1.2	328	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
9372	17.891479	172.16.41.135	52.111.224.7	TLSv1.2	2313	Application Data
9374	17.901531	52.111.224.7	172.16.41.135	TCP	60	443 → 56569 [ACK] Seq=9283 Ack=1783 Win=18688 Len=0
9375	17.901531	52.111.224.7	172.16.41.135	TCP	60	443 → 56569 [ACK] Seq=9283 Ack=2582 Win=21760 Len=0
9451	18.140473	172.16.41.135	52.110.14.146	TCP	66	56570 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9452	18.140826	172.16.41.135	52.110.14.146	TCP	66	56571 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9454	18.146720	52.110.14.146	172.16.41.135	TCP	66	443 → 56570 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256
9455	18.146720	52.110.14.146	172.16.41.135	TCP	66	443 → 56571 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256



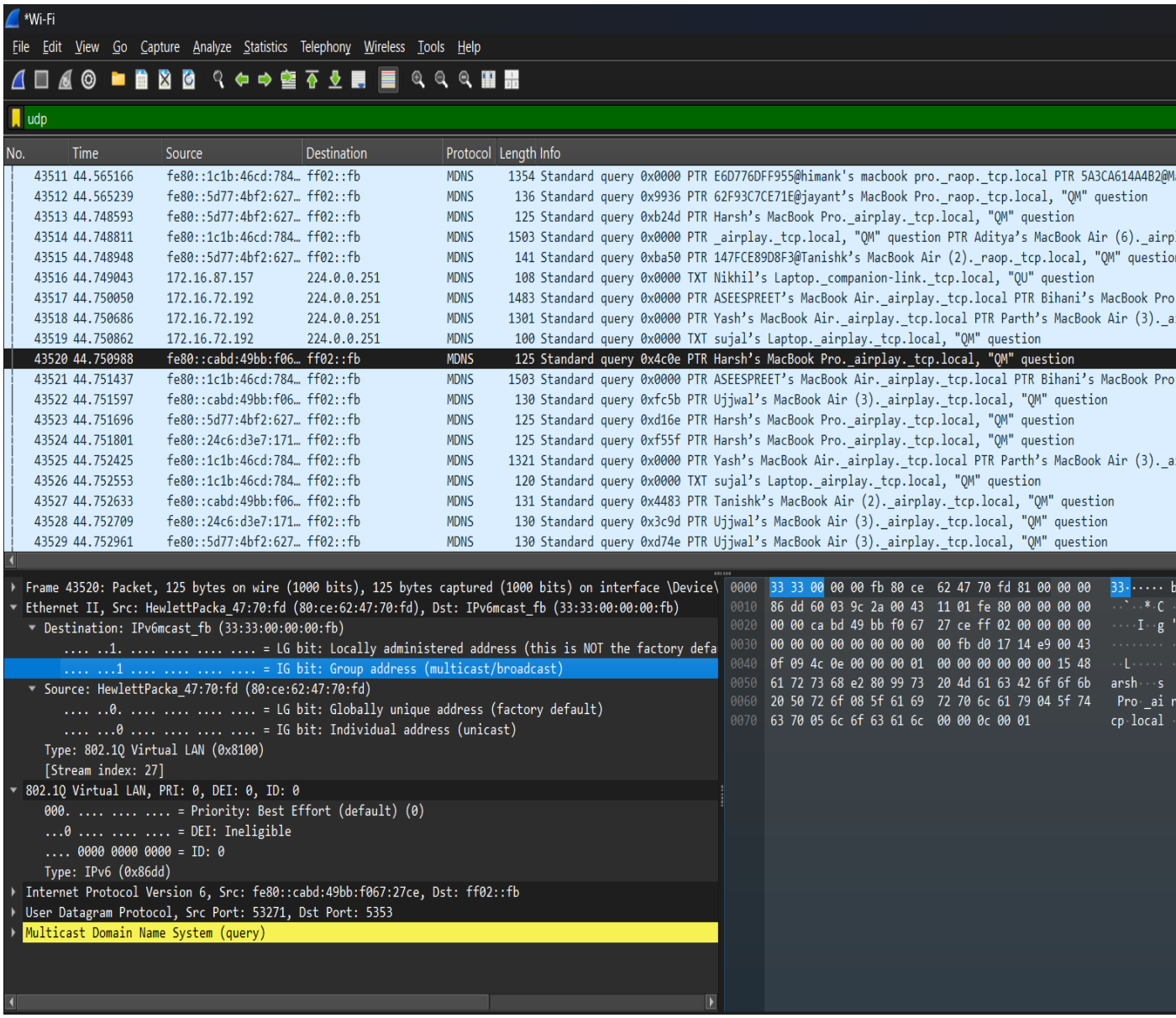
Wireshark · Packet 8976 · Wi-Fi

```
▸ Frame 8976: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id
▸ Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
▸ Internet Protocol Version 4, Src: 52.111.224.7, Dst: 172.16.41.135
▾ Transmission Control Protocol, Src Port: 443, Dst Port: 56569, Seq: 0, Ack: 1, Len: 0
    Source Port: 443
    Destination Port: 56569
    [Stream index: 2]
    [Stream Packet Number: 2]
  ▸ [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 1333957722
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 1272899674
    1000 .... = Header Length: 32 bytes (8)
  ▸ Flags: 0x012 (SYN, ACK)
    Window: 14600
    [Calculated window size: 14600]
    Checksum: 0x551f [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation (NOP), Window scale
  ▸ [Timestamps]
  ▸ [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]

0000  98 43 fa 0a fe 60 04 d5  90 6e 91 0f 08 00 45 00   ·C···`·· ·n····E·
0010  00 34 f9 1f 40 00 40 06  57 96 34 6f e0 07 ac 10   ·4··@·@· W·4o····
0020  29 87 01 bb dc f9 4f 82  94 5a 4b de e8 5a 80 12   )·····O· ·ZK··Z··
0030  39 08 55 1f 00 00 02 04  05 b4 01 01 04 02 01 03   9·U····· ********
0040  03 08                                              ··
```

No.: 8976 · Time: 17.345076 · Source: 52.111.224.7 · Destination: 172.16.41.135 · Protocol: TCP · Length: 66 · Info: 443 → 56569 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256

3. To capture and analyze UDP packets



4. To capture and analyze DNS query and response packets

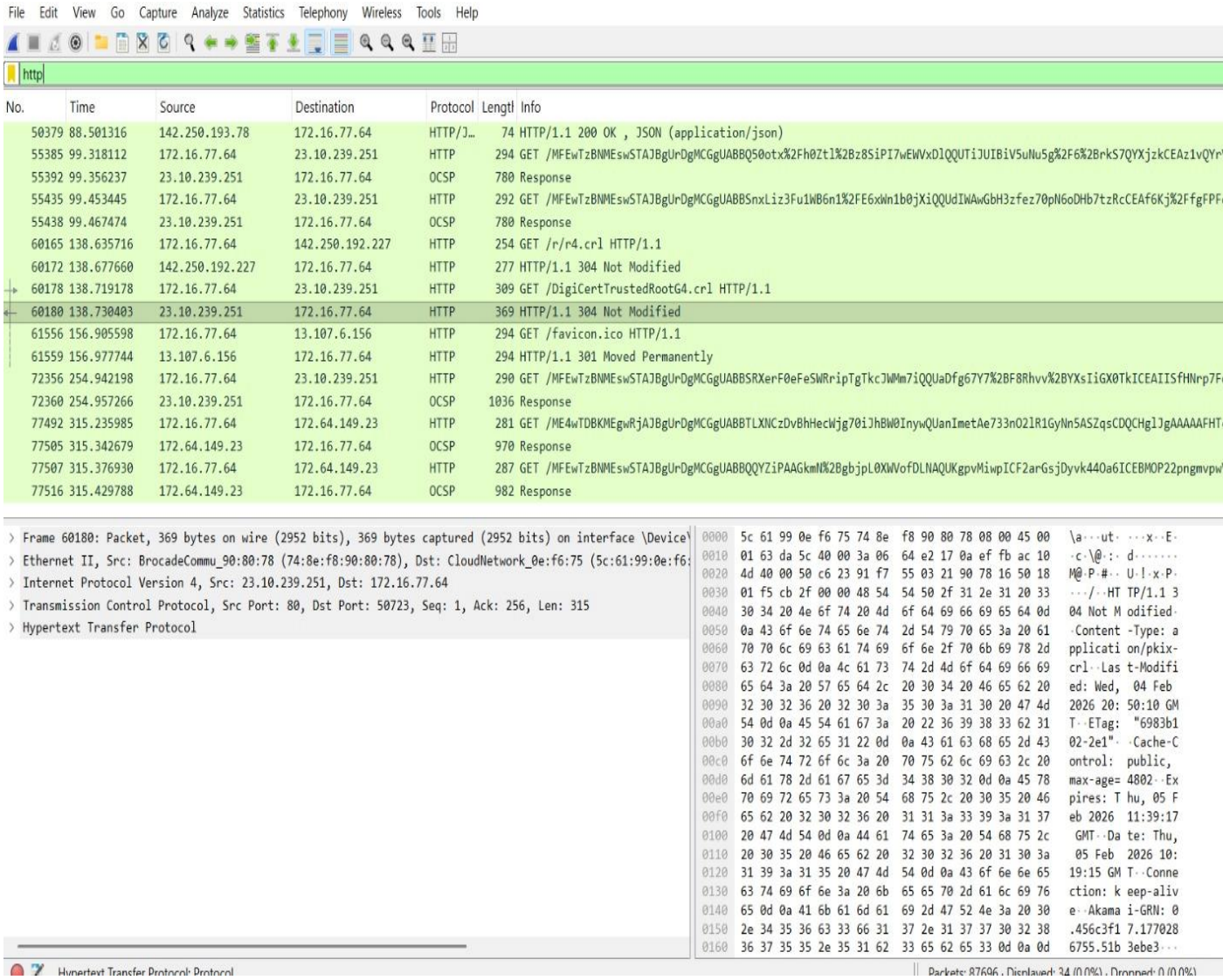
dns

No.	Time	Source	Destination	Protocol	Length	Info
71950	92.027577	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x0003 A fa000000191.resources.office.net
16846	31.105719	172.16.41.135	172.31.1.6	DNS	85	Standard query 0x03bf A fd.api.iris.microsoft.com
75735	92.889790	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x04e6 A fa000000193.resources.office.net
75773	92.908395	172.16.41.135	8.8.8.8	DNS	92	Standard query 0x04e6 A fa000000193.resources.office.net
17123	31.660998	172.16.41.135	172.31.1.6	DNS	78	Standard query 0x0ec8 AAAA uci.cdn.office.net
64804	90.185509	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x0eed A fa000000145.resources.office.net
66245	90.652437	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x15eb A fa000000175.resources.office.net
8967	17.321777	172.16.41.135	172.31.1.6	DNS	91	Standard query 0x1d09 A messaging.engagement.office.com
45655	85.668091	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x278d A fa000000128.resources.office.net
54196	87.491990	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x293c A fa000000136.resources.office.net
44264	85.236578	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x2c65 A fa000000124.resources.office.net
30496	57.952915	172.16.41.135	172.31.1.6	DNS	87	Standard query 0x2dd3 AAAA roaming.svc.cloud.microsoft
30521	57.983417	172.16.41.135	8.8.8.8	DNS	87	Standard query 0x2dd3 AAAA roaming.svc.cloud.microsoft
14265	27.005871	172.16.41.135	172.31.1.6	DNS	90	Standard query 0x2e5c AAAA self.events.data.microsoft.com
14264	27.005784	172.16.41.135	172.31.1.6	DNS	90	Standard query 0x31bd A self.events.data.microsoft.com
9492	18.257844	172.16.41.135	172.31.1.6	DNS	80	Standard query 0x32c2 A substrate.office.com
42619	82.146241	172.16.41.135	172.31.1.6	DNS	80	Standard query 0x3535 AAAA cs.dds.microsoft.com
13718	26.441137	172.16.41.135	172.31.1.6	DNS	76	Standard query 0x391e A fs.microsoft.com
66257	90.654212	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x3c3a AAAA fa000000156.resources.office.net
66249	90.652830	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x3c74 AAAA fa000000175.resources.office.net
13719	26.441236	172.16.41.135	172.31.1.6	DNS	76	Standard query 0x3ee1 AAAA fs.microsoft.com

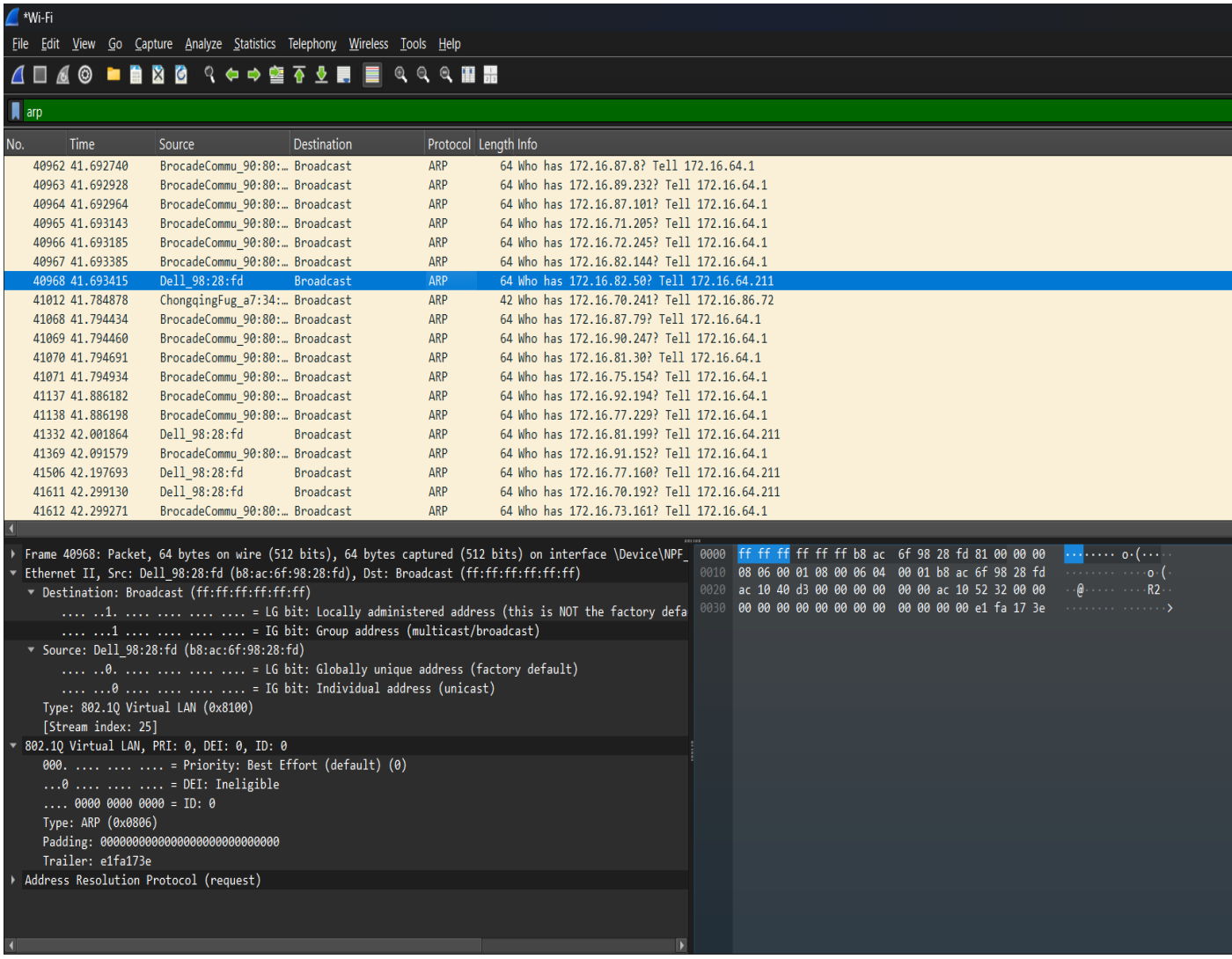
Wireshark · Packet 45655 · Wi-Fi

```
▸ Frame 45655: Packet, 92 bytes on wire (736 bits), 92 bytes captured (736 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, i
▾ Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
  ▸ Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
  ▸ Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
    Type: IPv4 (0x0800)
    [Stream index: 4]
▾ Internet Protocol Version 4, Src: 172.16.41.135, Dst: 172.31.1.6
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 78
    Identification: 0xffd5 (65493)
  ▸ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.16.41.135
    Destination Address: 172.31.1.6
    [Stream index: 176]
▾ User Datagram Protocol, Src Port: 64655, Dst Port: 53
    Source Port: 64655
    Destination Port: 53
    Length: 58
    Checksum: 0x8308 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 539]
    [Stream Packet Number: 23]
  ▸ [Timestamps]
    UDP payload (50 bytes)
▸ Domain Name System (query)
```

5. To capture and analyze HTTP packets



6. To analyze ARP request and reply packets



7. **To identify source and destination MAC and IP addresses in captured packets.**

tcp

No.	Time	Source	Destination	Protocol	Length	Info
6892	13.725480	172.16.41.135	13.204.106.34	TCP	55	60059 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
6896	13.728040	13.204.106.34	172.16.41.135	TCP	66	443 → 60059 [ACK] Seq=1 Ack=2 Win=103 Len=0 SLE=1 SRE=2
7705	14.957805	172.16.41.135	74.125.200.188	TCP	55	58464 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1
7746	15.038603	74.125.200.188	172.16.41.135	TCP	66	5228 → 58464 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2

```
▸ Frame 6892: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{
▾ Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
  ▸ Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
  ▸ Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
    Type: IPv4 (0x0800)
    [Stream index: 4]
▾ Internet Protocol Version 4, Src: 172.16.41.135, Dst: 13.204.106.34
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 41
    Identification: 0xba38 (47672)
  ▸ 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.16.41.135
    Destination Address: 13.204.106.34
    [Stream index: 213]
▸ Transmission Control Protocol, Src Port: 60059, Dst Port: 443, Seq: 1, Ack: 1, Len: 1
```

```
0000  04 d5 90 6e 91 0f 98 43  fa 0a fe 60
0010  00 29 ba 38 40 00 80 06  00 00 ac 10
0020  6a 22 ea 9b 01 bb bb 88  e8 46 c2 18
0030  00 fc 4d a1 00 00 00
```